3.3.1 (a) Show pullback is contravariant.

$$\varphi : V_1 \to V_2$$

$$\varphi^* : \Omega^{\otimes n}(V_2) \to \Omega^{\otimes n}(V_1)$$

$$f(q_2)(dq_2)^n \mapsto f(\varphi(q_1))\,(\varphi'(q_1))^n\,(dq_1)^n.$$

Pf: Let $\psi : V_2 \to V_3$

$\forall w_3 \in \Omega^{\otimes n}(V_3)$ $\quad w_3 = f(q_3)(dq_3)^n$.

$$\Rightarrow (\psi\circ\varphi)^*(f(q_3)(dq_3)^n) = f(\psi(\varphi(q_1)))\,(\psi'(\varphi(q_1)))^n\,(\varphi'(q_1))^n(dq_1)^n$$

$\Omega^{\otimes n}(V_3) \to \Omega^{\otimes}(V_1)$

$$= \varphi^*\left(f(\psi(q_2))\,(\psi'(q_2))^n(dq_2)^n\right) = \varphi^*\left(\psi^* f(q_3)(dq_3)^n\right).$$

(b) Show if $\iota : V_1 \hookrightarrow V_2$, then $\iota^* w = w|_{V_1}$ $\quad w \in \Omega^{\otimes n}(V_2)$.
be restriction

Pf: Let $w_2 = f(q_2)(dq_2)^n \in \Omega^{\otimes n}(V_2)$.

then $\iota^*(w_2) = f(\iota(q_1))\,(\iota'(q_1))^n\,(dq_1)^n = f(q_1)(dq_1)^n$.

$$= f(q_2)(dq_2)^n = w|_{V_1}.$$

(c) Show if φ holomorphic bijection of open sets in $\mathbb{C}$

then $(\varphi^{-1})^* = (\varphi^*)^{-1}$.

Pf: $\varphi : V_1 \xrightarrow{\sim} V_2$.

$$\begin{cases} (\varphi^*)^{-1}(f(q_2)(dq_2)^n) = \left(f(\varphi(q_1))\,(\varphi'(q_1))^n(dq_1)^n\right)^{-1} \\ (\varphi^{-1})^*(f(q_2)(dq_2)^n) = \end{cases}$$

From (a), $(\varphi^{-1}\circ\varphi)^* = \varphi^*\circ(\varphi^{-1})^*$

$$\overset{\shortparallel}{(\mathbb{1}_V^*)^*} = \mathbb{1}_{\Omega^{\otimes n}(V)}.$$

Thus $\varphi^*\circ(\varphi^{-1})^* = \mathbb{1}_{\Omega^{\otimes n}(V)}$. $\quad (\varphi^*)^{-1} = (\varphi^{-1})^*$ on $\Omega^{\otimes n}(V)$

(3.3.1) (d) i) Show if $\pi: V_1 \to V_2$ holomorphic surjection of open sets in $\mathbb{C}$, then π^* is injection.

Pf: $\pi^*(f(q_2)(dq_2)^n) = f(\pi(q_1))(\pi'(q_1))^n (dq_1)^n = 0$ $\forall q_1$ ~~$\forall q_1$~~.

$\pi(q_1)$ runs over V_2 tells

Since $(dq_1)^n \neq 0$, $\pi'(q_1) \neq 0$, we have $f(\pi(q_1)) = 0$.

Hence $f = 0$, which gives $f(q_2)(dq_2)^n = 0$.

ii) If $i: V_1 \to V_2$ injection of open sets in $\mathbb{C}$ need i^* be a surjection?

Sol: No e.g. $f(q_1)(dq_1)^n = \dfrac{dq_1}{q_1} \in \Omega^1(\mathbb{C}^\times)$.

$i: \mathbb{C}^\times \hookrightarrow \mathbb{C}$.

There are no elements $f(q_2)(dq_2) \in \Omega^1(\mathbb{C})$.

$$i^*(f(q_2)(dq_2)) = \frac{i'(q_1)\,dq_1}{i(q_1)} = \frac{dq_1}{q_1}.$$

3.3.2 (a) Let $\psi = \rho \circ \delta$, $\delta \in SL_2(\mathbb{Z})$, $\rho(z) = e^{2\pi i z/h}$

Let $w = g(q)(dq)^n$. Show $\psi^*(w) = f(\tau)(d\tau)^n$

(δτ = z) with $f = \tilde{f}[\delta]_{2n}$, $\tilde{f}(z) = (2\pi i/h)^n q^n g(q)$, $q = e^{2\pi i z/h}$

Pf: $\psi^*(w) = g(\psi(\delta^{-1}(z)))(\psi'(\delta^{-1}(z)))^n (d_q \delta^{-1}(z))^n$.

By 3.3.1 (a), $\psi^* = (\rho \circ \delta)^* = \delta^* \circ \rho^*$.

$$\rho^*(w) = g(q)\left(\frac{2\pi i}{h} q\, dz\right)^n = \tilde{f}(z)(dz)^n.$$

Recall $d\delta(\tau) = \dfrac{d\tau}{(c\tau+d)^2}$

thus $\delta^*(\tilde{f}(z)(dz)^n) = \tilde{f}\left(\delta\tau\left(\dfrac{d\tau}{(c\tau+d)^2}\right)\right)^n$

$= \tilde{f}(\delta(\tau))(c\tau+d)^{-2n}(d\tau)^n = \tilde{f}[\delta]_{2n}(\tau)(d\tau)^n = f(\tau)(d\tau)^n$

(3.3.2) (b) Consider $\pi : \mathcal{H} \to X(\Gamma)$ as in diagram (3.5).

Show if the local meromorphic differentials w_j pull back under ψ_j^* to restrictions of some common ~~loco~~ global meromorphic differential form on $\mathcal{H}$ then they are compatible.

Pf: $\psi_j^*(w_j) = f(\tau)\, d\tau$. common on $\mathcal{H}$.

$\psi_j : U_j \to \mathcal{H}$.

$$\begin{tikzcd} & U_j \cap U_k \arrow[dl, "\psi_j"'] \arrow[d] \arrow[dr, "\psi_k"] & \\ V_j \arrow[r, "\varphi_j^{-1}"'] & \pi(U_j \cap U_k) \arrow[r, "\varphi_k"'] & V_k \end{tikzcd}$$

On overlap $U_i \cap U_j$, $\varphi_i = \gamma_{ij} \circ \psi_j$ for some $\gamma_{ij} \in \Gamma$

Then $\psi_i^*(w_i) = (\gamma_{ij} \circ \psi_j)^*(w_i) = \psi_j^* \gamma_{ij}^*(w_i)$.

$\psi_i^*(w_i) = f(\tau)(d\tau) = \psi_j^*(w_j)$.

$\Rightarrow \psi_j^*(\gamma_{ij}^*(w_j) - w_j) = 0$.

By 3.3.1 (d), pullback by holomorphic surjection ψ_j is injective

Therefore $\gamma_{ij}^*(w_i) = w_j$ on $U_i \cap U_j$.

3.3.3. For $\gamma \in GL_2(\mathbb{C})$, $\gamma'(\tau) = \dfrac{\det \gamma}{j(\gamma,\tau)^2}$. This generalizes Lem 1.2.2(e).

Pf: $\gamma(\tau) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tau = \dfrac{a\tau + b}{c\tau + d}$.

$$\gamma'(\tau) = \left(\frac{a\tau+b}{c\tau+d}\right)' = \frac{a(c\tau+d) - (a\tau+b)c}{(c\tau+d)^2} = \frac{ad-bc}{j(\gamma,\tau)^2} = \frac{\det\gamma}{j(\gamma,\tau)^2}.$$

3.3.4 (a) For $U_j \subset \mathcal{H}$, show local differential w_j in (3.6) $w_j = \frac{g_j(q)}{(hq)^n}(dq)^n$ pulls back under ρ_j to λ_j.

How was w_j found in the first place?

Pf: In U_j, $q = e^{2\pi i\tau/h}$. $dq = \frac{2\pi i}{h} q\, d\tau$.

$$\Rightarrow d\tau = \frac{h}{2\pi i}\frac{dq}{q}.$$

$$w_j = \frac{g_j(q)}{(hq)^n}\left(\frac{2\pi i}{h} q\, d\tau\right)^n = (2\pi i)^n \frac{g_j(q)}{h^{2n}}(d\tau)^n.$$

Let $f(\tau) = (2\pi i)^n g_j(q)$.

then $\lambda_j = (2\pi i)^n g_j(q)\left(\frac{h}{2\pi i}\right)^n \frac{(dq)^n}{q^n} = g_j(q)\frac{(dq)^n}{(hq)^n}$.

How found:

$\lambda_j : \tau \mapsto \tau + h$. $d\tau \sim \frac{dq}{hq}$

$$\Rightarrow dq \mapsto \frac{dq}{hq}.$$

(b) Same question for w_j in (3.7) $w_j = \frac{g_j(q)}{(\frac{2\pi i}{h}q)^n}(dq)^n$.

$d\tau = \frac{h}{2\pi i}\frac{dq}{q}$. holomorphic at $q=0$.

$$\lambda_j = f(\tau)(d\tau)^n = g_j(q)\frac{(dq)^n}{\left(\frac{2\pi i}{h}q\right)^n}.$$

$$\rho^*(w_j) = \lambda_j.$$

3.3.5 Prove formulas

(3.8) $\nu_0(w_j) := \nu_0\left(\frac{g_j(q)}{(hq)^{k/2}}\right) = \nu_{\tau_j}(f) - \frac{k}{2}\left(1 - \frac{1}{h}\right)$.

Pf: Let $\tau_j \in \mathcal{H}$ be elliptic point with period h.

$$\left.\begin{array}{l} z = \tau - \tau_j \\ q = z^h \end{array}\right\} \Rightarrow \nu_0(g_j(q)) = \frac{1}{h}\nu_{\tau_j}(f).$$

$$w_j = g_j(q)\frac{(dq)^n}{(hq)^n} = g_j(q)\frac{h^n z^{n(h-1)}(dz)^n}{h^n z^{hn}} = g_j(q) z^{-n}(dz)^n$$

(3.3.5) $$\nu_0(\omega_j) = \nu_0(g_j(z^h)) - n = \frac{1}{h}\nu_{\tau_j}(f) - n.$$
$$= \nu_{\pi(\tau_j)}(f) - n.$$

(3.9) $$\nu_0(\omega_j) := \nu_0 \frac{g_j(q)}{(2\pi i q/h)^{k/2}} = \nu_{\pi(s_j)}(f) - \frac{k}{2}.$$

Pf: $$\omega_j = g_j(q)\frac{(dq)^n}{\left(\frac{2\pi i}{h}q\right)^n} = g_j(z^h)\frac{h^n z^{n(h-1)}(dz)^n}{(2\pi i/h)^n z^{hn}}$$
$$= g_j(z^h)\, z^{-n}(dz)^n \cdot \left(\frac{h^2}{2\pi i}\right)^n$$
$$\nu_0(\omega_j) = \nu_0(g_j(z^h)) - n = \frac{1}{h}\nu_{\tau_j}(f) - n$$
$$= \nu_{\pi(\tau_j)}(f) - \frac{k}{2}$$

3.3.6 Prove $S_2(\Gamma) \xrightarrow{\sim} \Omega^1_{hol}(X(\Gamma))$ as $\mathbb{C}$-vector spaces.

Show in proof elements of $S_2(\Gamma)$ vanish at elliptic points of $X(\Gamma)$. argue directly by examining text ~~before~~ leading up to (3.6).

Pf: $q = e^{2\pi i\tau/h}$ $\quad d\tau = \frac{h}{2\pi i}\frac{dq}{q}$. $\quad f(\tau) = \sum_{n\geq 1} a_n q^n$

Cusp:
$$\Rightarrow f(\tau)d\tau = \frac{h}{2\pi i}\left(\sum_{n\geq 1} a_n q^{n-1}\right)dq \quad \text{holomorphic at } q=0.$$

Elliptic points: $z = \tau - \tau_0$ $\quad w = z^h$.
$z \mapsto \mu_h z$. $\quad f(\tau) = \sum_{m\geq 0} a_m z^m$

Γ-invariance $\Rightarrow f(\mu_h z)\, d(\mu_h z) = f(z)\, dz$.
$$\Rightarrow \mu_h f(\mu_h z) = f(z) \Rightarrow a_m(\mu_h^{m+1} - 1) = 0 \Rightarrow m+1 \equiv 0 \ (h)$$
hence $f(z) = z^{h-1}$ holomorphic
$$f(\tau)d\tau = z^{h-1}dz = \frac{1}{h}dw.$$

Now let $\omega \in \Omega^1_{hol}(X(\Gamma))$ $\quad \pi^*(\omega) = f(\tau)d\tau$.
Γ-invariance $\Rightarrow k=2$, holomorphy at cusps $\Rightarrow$ vanish at cusps.